Lesson 12: Practical Strategies for Psychological Measurement

English Stage 4 (Psychology) · Unit 2: Research Ethics and Measurement

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 4.4 Practical Strategies for Psychological Measurement; 4.5 Psychological Measurement (Summary).

In this lesson

You learn the four steps of measuring a construct in a real study: defining it conceptually, defining it operationally (with an existing or a new measure), implementing the measure and evaluating it. You practise hedged recommendations and the passive voice, and you write the Measures part of your proposal.

Learning goals

After this lesson you should be able to:

- Describe the four steps of the measurement process.
- Decide whether to use an existing measure or create a new one, and say where to find measures.
- Explain how to reduce participant reactivity.
- Describe measures and procedures with the passive voice and hedged recommendations.

Key language

Term or phrase	Meaning	Example
proprietary	owned by a publisher, so you must buy it	the Beck Depression Inventory and many intelligence tests
PsycTESTS	an APA collection of measures used in previous research	The APA also publishes *PsycTESTS*.
pilot test	A small-scale study conducted primarily to be sure that a procedure works as planned.	Test several people before large-scale data collection.
participant reactivity	people's reactions to being measured that reduce reliability and validity	Agreeable participants might respond in ways they believe they are expected to.
socially desirable responding	doing or saying things because participants think it is the socially appropriate thing	agreeing they are a person of worth so as not to look bad
demand characteristics	Features of a study that cue participants as to how the researcher expects them to behave.	reading about heart disease just before an exercise attitude measure
blind	unaware of the measure's intent or of the hypothesis	a helper who is "blind" to the hypothesis

Term or phrase	Meaning	Example
standardize	make every interaction the same for all participants	always reading the same instructions word for word
strive for	try hard to achieve	You should strive for simplicity.
back to the drawing board	start the planning again	revise the measure, the definition or the manipulation

Rules and explanations

The ideas

Step 1: Define the construct conceptually. "Memory" is too vague, because psychologists see memory as a set of semi-independent systems. For **long-term episodic memory** (memory for previous experiences), asking participants to recall a word list learned last week makes sense; asking them to remember to do a task in the future does not. Read the literature to see how others define the construct.

Step 2: Define it operationally, turning the concept into something observable (stress: a Perceived Stress Scale score, cortisol in saliva or the number of recent stressful life events).

- **Use an existing measure** when possible: you save time, there is already evidence of validity, and your results can be compared with earlier ones. Choose the most common, the best supported, the best fit for your aspect of the construct, or the easiest (the 10-item TIPI measures the Big Five less reliably but quickly). Find measures in published articles, the APA's *Directory of Unpublished Experimental Measures* and *PsycTESTS*; proprietary tests are described in *Tests in Print* and the *Mental Measurements Yearbook*.
- **Create your own** when none exists, existing ones are too difficult, or you want to test convergent validity. Most new measures adapt old ones: the Stroop task (naming the colours words are printed in) was adapted for social anxiety: socially anxious people name the colour of words like "stupid" more slowly. Keep it simple (clear instructions, practice items, time for questions, brevity) but use **multiple items**: they cover the construct better (content validity) and irrelevant factors cancel out (reliability). Items must combine into one score: income, credit score and a "thrifty" rating cannot be added up; ratings of 10 statements on the same five-point scale can. Test several people first.

Step 3: Implement the measure under similar, quiet conditions, and reduce **reactivity**:

Problem	Precaution from the book
frustration	make the procedure clear and brief
socially desirable responding	guarantee anonymity: seat people apart, give everyone the same pen (no pink glitter pen), use envelopes or a drop box
demand characteristics	do not reveal the hypothesis ("Money Questionnaire", not "Are You Financially Responsible?")
researcher expectations	use a helper who is blind to the hypothesis; standardize all interactions

Step 4: Evaluate the measure. Do not assume it worked for your sample. Check test-retest reliability if people are tested twice, internal consistency with a split-half correlation or Cronbach's α, and criterion validity through correlations with other measures. A successful manipulation is also evidence: MacDonald and Martineau's mood measure showed a clear difference between the positive- and negative-thought groups, showing that both the manipulation and the measure worked. If your data cast doubt on the measure, ask why (the measure, the conceptual definition or the manipulation) and go back to the drawing board.

Language focus

1. Hedged recommendations. The book gives advice without sounding like a rulebook:

- "**It is usually a good idea to** use an existing measure that has been used successfully in previous research."
- "**It is nearly always better for** a measure **to** include multiple items rather than a single item."
- "**Be aware** also **that** people can react in a variety of ways to being measured that reduce the reliability and validity of the scores."

2. The passive voice. The passive puts the measure or the participants first and leaves out who acts, as a Method section usually does:

- "stress **can be operationally defined** as people's scores on a stress scale"
- "Participants **are** often **tested** in groups because it is efficient"
- "the effects of your expectations **can be minimized** by arranging to have the measure administered by a helper" In a proposal, use the future passive: "Memory **will be measured** with ..."

> **Note for Arabic speakers** - Do not drop *be* in the passive: "Participants will **be** tested", not "Participants will tested". - *Research*, *advice* and *information* are uncountable: "much research", "a piece of advice", not "researches" or "an advice".

Practice

1. Reading questions (section 4.4)

1. What are the four broad steps in the measurement process?
2. Why must a researcher who wants to measure "memory" be more precise?
3. What are the three advantages of using an existing measure?
4. Why is it nearly always better to use multiple items, and why could the student's three questions about financial responsibility not form a multiple-item measure?
5. How did MacDonald and Martineau's mood measure provide evidence of validity?

2. Book exercises (4.5; they cover the whole chapter, Lessons 10-12)

1. Practice: Complete the Rosenberg Self-Esteem Scale and compute your overall score. (The scale is in section 4.1, Lesson 10.)

2. Practice: Think of three operational definitions for sexual jealousy, decisiveness, and social anxiety. Consider the possibility of self-report, behavioral, and physiological measures. Be as precise as you can.
3. Practice: For each of the following variables, decide which level of measurement is being used. a. A university instructor measures the time it takes her students to finish an exam by looking through the stack of exams at the end. She assigns the one on the bottom a score of 1, the one on top of that a 2, and so on. b. A researcher accesses her participants' medical records and counts the number of times they have seen a doctor in the past year. c. Participants in a research study are asked whether they are right-handed or left-handed.
4. Practice: Ask several friends to complete the Rosenberg Self-Esteem Scale. Then assess its internal consistency by making a scatterplot to show the split-half correlation (even- vs. odd-numbered items). Compute the correlation coefficient too if you know how. (Self-study version: plan how you would do this, then use the invented data below.)

Friend (invented data for practice)	A	B	C	D	E	F
Odd-numbered items (0-15)	12	9	14	7	10	5
Even-numbered items (0-15)	13	8	13	8	11	6

5. Discussion: Think back to the last college exam you took and think of the exam as a psychological measure. What construct do you think it was intended to measure? Comment on its face and content validity. What data could you collect to assess its reliability and criterion validity?
6. Practice: Write your own conceptual definition of self-confidence, irritability, and athleticism.
7. Practice: Choose a construct (sexual jealousy, self-confidence, etc.) and find two measures of that construct in the research literature. If you were conducting your own study, which one (if either) would you use and why?

3. Language Rewrite in the future passive, starting with the words in bold.

1. We will test **participants** individually in a quiet room.
2. A helper who does not know the hypothesis will give **the memory test**.
3. We will read **the same instructions** word for word to every participant.

Speak with your tutor

- "In PSYC 110 Lecture 7, Professor Bloom distinguishes semantic from episodic memory, and recall from recognition. Help me decide which kind of memory my study measures, and how."
- "In PSYC 110 Lecture 13, Professor Bloom says personality tests use catch questions such as 'I have never done anything I am ashamed of'. Is this the problem the book calls socially desirable responding?"
- "I want to measure sleep with a questionnaire. Help me plan a PsycTESTS search and choose between the two measures I find, as in Exercise 7. Correct my justification."

Writing task

Write the **Measures** part of your proposal (sleep and memory in university students, or your own topic): 250-350 words, using your definitions (Lesson 10) and reliability and validity plan (Lesson 11).

- For each variable: the measure, cited as (Author, Year), or, if new, its items, response scale, instructions and practice items; what participants do; how it is scored; its level of measurement.
- Evidence of reliability and validity from earlier research, or how you will check it.
- Implementation: setting, standardized instructions, a blind helper if possible, anonymity, a neutral questionnaire title.
- The future passive ("will be measured") and at least one hedged recommendation.

If you work on your own topic, describe your own measures in the same way.

Self-editing checklist:

- Could another researcher repeat my measurement from this description?
- Did I justify using an existing or a new measure?
- Does each multiple-item measure combine into one score?
- Did I include at least two precautions against reactivity?
- Are all passives formed with *be* + past participle?
- Did I cite each existing measure in APA style?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- PsycTESTS, the APA collection of measures (Exercise 7 and the Writing task): https://www.apa.org/pubs/databases/psyctests

Answer key

1. Reading questions

1. Conceptually defining the construct, operationally defining it, implementing the measure, and evaluating the measure.
2. Psychologists see memory as a set of semi-independent systems, so the researcher must say which kind (for example, long-term episodic memory) to choose a suitable task.
3. You save the time and trouble of creating your own; there is already some evidence of validity; your results can more easily be compared and combined with previous results.
4. Multiple items cover the construct more adequately (content validity), and summing or averaging lets irrelevant factors cancel out (reliability). Income, a credit score and a thrift rating cannot obviously be combined into one score.
5. After the manipulation, the mood measure showed a clear difference between the groups: evidence that the manipulation worked and that the measure was valid.

2. Book exercises

1. Items 1, 2, 4, 6 and 7 score 3, 2, 1, 0 from *Strongly Agree* to *Strongly Disagree*; Items 3, 5, 8, 9 and 10 score 0, 1, 2, 3. Totals range from 0 to 30; there is no "correct" score. Check: *Agree* to all ten items gives $5 \times 2 + 5 \times 1 = 15$.
2. (example answers) Sexual jealousy: self-report, total ratings (1-5) on statements such as "I would be upset if my partner flirted with someone"; behavioral, the number of questions about a third person that a participant asks a partner in a recorded 10-minute conversation; physiological, the rise in heart rate while imagining a partner flirting. Decisiveness: self-report, ratings of "I make decisions quickly"; behavioral, mean seconds to choose between two products over 20 trials; physiological, harder (skin conductance during choices, whose validity would need testing). Social anxiety: self-report, a questionnaire on fear in social situations; behavioral, Stroop colour-naming time for socially negative words; physiological, heart rate or galvanic skin response while giving a short speech.
3. a. Ordinal: the scores are ranks of finishing time, showing order but not how much longer. b. Ratio: a count with a true zero. c. Nominal: category labels with no order.
4. Plan: reverse-score Items 3, 5, 8, 9 and 10, total each friend's odd- and even-numbered items, plot one point per friend and compute the correlation. With the invented data the points rise from lower left to upper right and r = +.95: good internal consistency (+.80 or more).
5. (example answer) My last exam was meant to measure knowledge of research methods. Face validity was good, as every question was about course concepts; content validity was weaker, with many questions on ethics but few on measurement. For reliability I could compute a split-half correlation between odd- and even-numbered questions; for criterion validity I could correlate exam scores with later final-exam grades (predictive validity).
6. (example answers) Self-confidence: a general tendency to believe in one's ability to succeed and to act on it, for example by trying difficult tasks; related to, but narrower than, self-esteem.

Irritability: a tendency to become annoyed or angry easily at minor frustrations, with negative feelings and impatient behavior. Athleticism: the ability to perform physical tasks requiring strength, speed, endurance and coordination; it should correlate with performance across sports.

7. (example answer) For stress: the Perceived Stress Scale (Cohen, Kamarck, & Mermelstein, 1983) and the Social Readjustment Rating Scale (Holmes & Rahe, 1967). For students' stress before exams I would use the Perceived Stress Scale, because it measures present feelings of stress, while the other counts major life events over a whole year. (Search for measures in PsycTESTS.)

3. Language

1. Participants will be tested individually in a quiet room. 2. The memory test will be given (administered) by a helper who does not know the hypothesis. 3. The same instructions will be read word for word to every participant.

My notes